

Experiment 6

AIM: setup Nginx to handle proxy requests and load balancing.

step 1: Create a Project

```
C:\> mkdir load-balance-demo
```

```
C:\> cd load-balance-demo
```

```
C:\load-balance-demo>npm init -y
```

Wrote to C:\load-balance-demo:

```
{
  "name": "users",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}
```

(2): Create server.js

```
const express = require("express");
const app = express();
const PORT = process.env.PORT || 3000;
app.get("/", (req, res) => {
  res.send(`Response from Server running on port ${PORT}`);
});
app.listen(PORT, () => {
  console.log(`Server running on port ${PORT}`);
});
```

(3) C:\load-balance-demo>node index.js

Server 1 running on port 3000

(4) Create a Folder Load balancing

Add node.js project

change index.js file (server - 1)

port: 3000

```
const express = require("express");
const app = express();
const PORT = 3000;

app.get("/", (req, res) => {
  setTimeout(() => {
    res.send("Slow Server - Port 3000");
  }, 5000);
});
```

```
app.listen(PORT, () => {  
  console.log(`Server 1 running on port ${PORT}`);  
});
```

Add node.js project
change index.js file (server - 2)
port : 3001

```
const express = require("express");  
const app = express();  
const PORT = 3001;  
  
app.get("/", (req, res) => {  
  res.send("Fast Server - Port 3001");  
});  
  
app.listen(PORT, () => {  
  console.log(`Server 2 running on port ${PORT}`);  
});
```

Step 2: Run Multiple Servers

Open 2 terminals:

Terminal 1:

At PORT=3000

node server.js

C:\load-balance-demo\load-balancing\server1>node index.js

Server 1 running on port 3000

Terminal 2:

at PORT=3001

node server.js

C:\load-balance-demo\load-balancing\server2>node index.js

Server 1 running on port 3001

Step 3. Configure NGINX for Load Balancing (Open nginx.conf and Modify http block)

```
http {  
  upstream myapp {  
    server localhost:3000;  
    server localhost:3001;  
  }  
  server {  
    listen 8080;  
    server_name localhost;  
    location / {  
      proxy_pass http://myapp;  
    }  
  }  
}
```

Step 4: Restart NGINX

```
C:\load-balance-demo>nginx -s stop
```

```
C:\load-balance-demo>Nginx
```

Now open:

<http://localhost:8080>

Refresh multiple times.

You can check:

Response from Server running on port 3000

Response from Server running on port 3001

Modify nginx.conf file:

```
upstream myapp {  
    least_conn;  
    server localhost:3000;  
    server localhost:3001;  
}
```

```
C:\load-balance-demo\load-balancing\server1>node index.js
```

```
Server 1 running on port 3000
```

```
C:\load-balance-demo\load-balancing\server2>node index.js
```

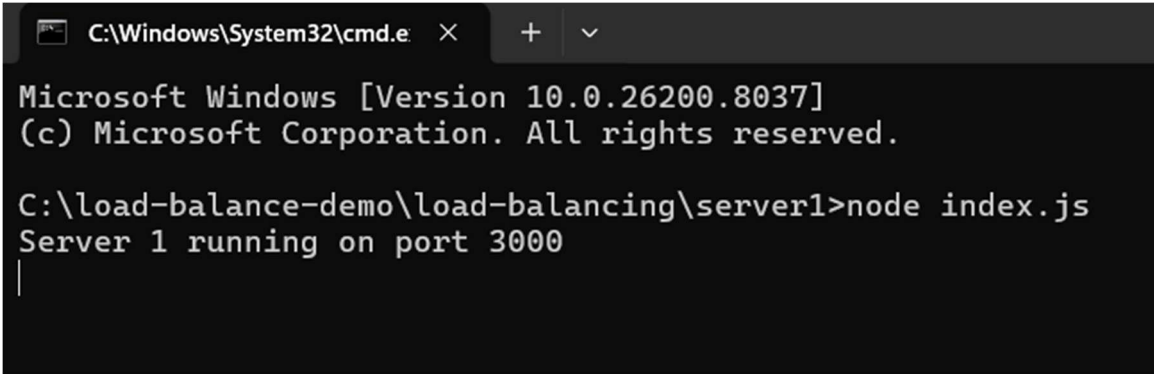
```
Server 1 running on port 3001
```

```
C:\load-balance-demo>nginx -t
```

```
nginx: the configuration file C:\load-balance-demo\nginx.conf syntax is  
ok
```

```
nginx: configuration file C:\load-balance-demo\nginx.conf test is  
successful
```

- **Output:**



```
C:\Windows\System32\cmd.e  X  +  v  
Microsoft Windows [Version 10.0.26200.8037]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\load-balance-demo\load-balancing\server1>node index.js  
Server 1 running on port 3000  
|
```

```
C:\Windows\System32\cmd.e  X  +  v

Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\load-balance-demo\load-balancing\server2>node index.js
Server 2 running on port 3001
|
```

```
C:\Windows\System32\cmd.e  X  +  v

Microsoft Windows [Version 10.0.26200.8037]
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C:\load-balance-demo>nginx -t
nginx: the configuration file C:\load-balance-demo/conf/nginx.conf syntax is ok
nginx: configuration file C:\load-balance-demo/conf/nginx.conf test is successful

C:\load-balance-demo>|
```

